

PX-TT-Open-WBO-Inc: Adding Partition Crossover to a MaxSAT solver

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Abstract—Partition Crossover is an efficient recombination operator designed in the context of gray-box optimization to be included in metaheuristics. It has shown to speed up the search in pseudo-Boolean optimization problems with low epistasis, including MaxSAT. The solver we propose here, is a variant of the TT-Open-WBO-Inc solver proposed by Alexander Nadel in the previous competitions. We included Partition Crossover in several algorithms inside the solver.

Index Terms—Partition crossover, gray-box optimization, TT-Open-WBO-Inc solver, MaxSAT Evaluation 2026

I. INTRODUCTION

We provide in this document a brief description of the proposed solver, called PX-TT-Open-WBO-Inc, which is based on TT-Open-WBO-Inc, by Alexander Nadel. The main novelty in our solver is the inclusion of Partition Crossover (PX), an efficient recombination operator for metaheuristic algorithms, which has been successfully included in single-objective search algorithms like DRILS [1] and, more recently, in multi-objective search algorithms [2]. In Section II, we describe this operator while in Section III we explain where was this operator included in the TT-Open-WBO-Inc solver.

II. PARTITION CROSSOVER

Before explaining how PX works, we need to describe the notation we use. The set $\mathbb{B}^n$ contains all the binary strings of length n . Given a set of variables Y , the notation 1_Y^n is used to represent a binary string of length n with 1 in the positions of the variables in Y and zero in the rest, for example, $1_{\{0,2\}}^3 = 101$. If the set Y has only one element, we will omit the curly brackets in the subscript, e.g., $1_3^4 = 1_{\{3\}}^4 = 0001$. Observe that the first index of the binary strings is 0. When the length is clear from the context we omit n . The operator $\oplus$ is a bitwise exclusive OR.

The *variable interaction graph* (VIG) [3] of a pseudo-Boolean function f is a graph $G = (V, E)$, where V is the set of variables and E is the set of edges representing all pairs of variables (x_i, x_j) having *nonlinear interactions*. We say that variables x_i and x_j have a nonlinear interaction when the expression $\Delta_i f(x) = f(x \oplus 1_i) - f(x)$ does depend

on x_j . Checking the dependency of $\Delta_i f(x)$ on x_j can be computationally expensive in general, but it is very fast in the case of MaxSAT formulas in CNF [4].

Let us now assume that we have two solutions, also called *parent* solutions: the *red* and *blue* solutions. We want to generate a new solution from them, called the *offspring* solution. All the variables with the same value in both solutions will also share the same value in the offspring. We remove from the VIG all the variables (vertices) that have the same “shared variable assignments” and also remove all edges that are incident on the vertices corresponding to these variables. This produces the so-called *recombination graph*.

The recombination graph also defines a reduced evaluation function. This new evaluation function is linearly separable, and decomposes into q subfunctions defined over the connected components of the recombination graph. Every recombination graph with q connected components induces a new *separable* function $g(x')$ that is defined as:

$$g(x') = a + \sum_{i=1}^q g_i(x'). \quad (1)$$

Partition crossover (PX) [5], generates an offspring, when recombining two parents, based on the recombination graph: all of the variables in the same recombining component in the recombination graph are inherited together from one of the two parents. PX selects the decision variables from one or another parent yielding the best partial evaluation for each subfunction $g_i(x')$. This way, PX obtains a best offspring among 2^q in $O(q)$ time, which is a remarkable result. The efficiency of PX depends on the number of connected components q in the recombination graph. Larger values for q provide a better performance. This number can be remarkably large in MaxSAT instances. Chen and Whitley [6] showed that $q = 1087$ in a recombination graph of a real SAT instance of the 2014 Competition¹. They also showed that PX can improve MaxSAT solvers by jumping between plateaus [7].

III. PX IN THE SOLVER

We modified the TT-Open-WBO-Inc solver by Alexander Nadel [8] and added PX in some of the algorithms inside the

We thank Darrell Whitley, the original inventor of Partition Crossover, and Renato Tinós, the main author of the PX for pseudo-Boolean functions, all the discussion about the use of PX and other operators in MaxSAT.

¹<http://www.satcompetition.org/2014/>

solver. In particular, we included PX in the NUWLS local search [9], MrsBeaver [10], LSU [11], [12], and LinearSU clustering [8]. We describe the way PX was included in more detail below. The submitted version of our solver uses IntelSAT [13].

In the case of NUWLS, whenever a solution better than the current one is found during the successive iterations, it is added to a list of elite solutions together with its cost, and an improvement flag is activated. This makes it possible to identify when the search is progressing correctly and when it has become trapped in a local optimum. Every 500 iterations, it is checked whether the solution list is non-empty and whether the improvement flag remains activated. If both conditions are satisfied, a recombination step using PX is performed in an attempt to improve the current solution. To do so, the list of obtained solutions is sorted, and the solution with the lowest cost is selected as the first parent. For the second parent, a solution with at least more than 0.5% differing bits is chosen. Among all candidate solutions satisfying this condition, the one with the smallest Hamming distance from the first parent is selected.

PX has been applied in MrsBeaver to avoid stagnation during the search for optimal solutions. Its operation is the same as the one described for NUWLS, but in this case the algorithm evaluates at each iteration whether there have been changes in the best solution found so far. This reference point is chosen according to the strategy-switching mechanism followed by the search algorithms, since they alternate every two iterations between conservative and aggressive local search. PX has also been implemented in LSU in order to escape from local optima whenever the search becomes stagnated.

In the case of LinearSU clustering, PX has been incorporated into the iterative search phase, where this operator will be applied whenever the current solution cannot be improved. The parent with the lowest cost from the list of candidate solutions will be selected as the first parent, while the second parent must differ by at least 0.5% of its bits. Unlike in previous applications of PX, the second parent is randomly selected from a list of five possible candidates. These candidate solutions are those that satisfy the condition of having a minimum percentage of differing bits with respect to the previously selected first parent. If the resulting solution has a lower cost than the second parent, it is also stored in the candidate list even if it does not improve the global solution. This is done to reduce the cost of future solutions obtained through recombination.

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